Troubleshooting Docker network issues

In this train, we'll develop the ability to identify and troubleshoot common network problems within Docker, using command-line tools for diagnosis.

Learning objectives

- Understand common network problems in Docker and how to troubleshoot using command-line tools.

Common Docker Network Issues

Some common docker networking issues you may encounter are:

- Containers can't talk to each other.
- Containers can't connect to the internet.
- You can't access an application running in a container from your computer.

We'll use the following command-line tools to investigate these issues:

- `docker network ls` - Shows you all the Docker networks.
- `docker network inspect network_name` - Gives you detailed information about a network.
- `docker ps` - Lists running containers.
- `docker inspect container_name_or_id` - Shows you a container's settings.
- `docker exec –it container_name_or_id /bin/bash` - Lets you run commands inside a container.
- `ping`, `netstat`, `ip route`, `nslookup` - Classic network tools you can use inside containers.

Containers can't communicate:

Containers that should be able to exchange data over the network are unable to do so. This could be due to incorrect network configurations, firewall issues, or application-level problems.

- Check the network:
    1. `docker network ls` : Make sure the containers are on the same network.
    2. `docker network inspect network_name` : check the subnet and IPAM settings.
- Check container IPs:
    1. `docker inspect container_name_or_id` : Find the containers' IP addresses.
    2. `docker exec –it container_name_or_id /bin/bash` : Enter a container's shell. `ping other_container_ip` : In the shell, try pinging the other container from this container.
- Check application ports:
    1. Make sure the application in the other container is listening on the correct port.
    2. `Use netstat –tuln` inside the container to see which ports are open.

Containers can't connect to the internet:

This indicates that containers lack the ability to reach external websites or services. This is often related to DNS resolution problems, routing issues, or network driver configurations.

- Check DNS
    1. `docker exec –it container_name_or_id /bin/bash` : Enter a container.
    2. `ping google.com` : Try pinging a website.
    3. `nslookup google.com` : Test DNS resolution.
    4. `cat /etc/resolv.conf` : Check the DNS settings inside the container.
- Check routing
    1. `ip route` : See the container's routing table.
    2. Make sure there's a default route to the internet.

Can't access an application in a container on your computer:

This occurs when you attempt to connect to an application exposed by a container (e.g., a web server) but the connection fails. This is often related to port mapping errors, firewall restrictions, or the application not running correctly.

- Check port mapping:
    1. `docker port` - See which ports are mapped.
    2. Make sure the port is mapped correctly.
- Check firewall:
    1. Check your firewall settings. Your computer's firewall might be blocking the connection.
- Check application:
    1. Ensure the application inside the container is running and listening on the mapped port.

Conclusion

You should now be able to effectively identify and fix problems within your Docker environment.

Key Takeaways

- We looked at common networking issues.
- We investigated how to diagnose problems using command-line tools.